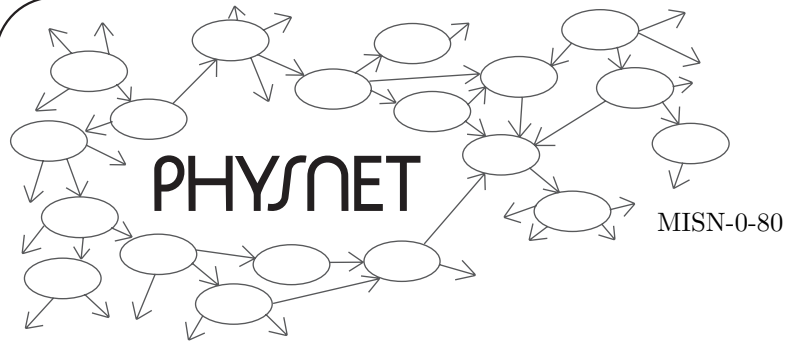
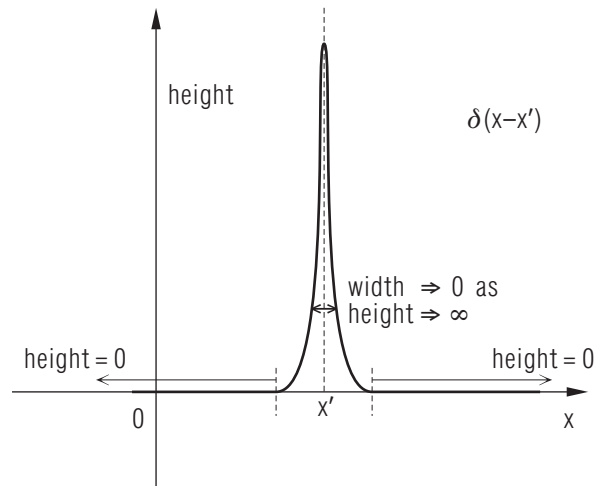




THE DELTA FUNCTION



THE DELTA FUNCTION
by
Jim O'Connell

Title: **The Delta Function**

Author: Jim O'Connell, Michigan State University

Version: 8/7/75

Evaluation: Stage 0

Length: 1 hr; 5 pages

Input Skills:

1. Knowledge of Coulomb's Law.
2. Understanding of the Kronecker delta operator.
3. Requires knowledge of Poisson's equations for a point charge, the Hermite polynomials, Parseval's relation, and the quantum mechanical position and momentum operation and their associated wave functions.

Output Skills (Knowledge):

- K1. List the properties of the delta function as a distribution and in its uses in integral calculus.
- K2. Show the relationship between the delta function and the Heaviside distribution.
- K3. Show the relationship between the Kronecker delta and the Dirac delta function.
- K4. Derive the momentum distribution in quantum mechanics.

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OF PROJECT PHYSNET

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Our publications are designed: (i) to be updated quickly in response to field tests and new scientific developments; (ii) to be used in both classroom and professional settings; (iii) to show the prerequisite dependencies existing among the various chunks of physics knowledge and skill, as a guide both to mental organization and to use of the materials; and (iv) to be adapted quickly to specific user needs ranging from single-skill instruction to complete custom textbooks.

New authors, reviewers and field testers are welcome.

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Views expressed in a module are those of the module author(s) and are not necessarily those of other project participants.

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